



Can expert opinion improve effort
predictions when exploiting cross-company
datasets?

- A case study in a small/medium company

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Outline

Motivation

- Motivation
 - Focus of the paper and our proposal

Study Design

- Empirical study
 - Design
 - Cross-Company Dataset Selection
 - Model Construction and Validation
 - Discussion

Study Results

Final remarks

- Final remarks and future work

Context

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- Effort estimation is a key management activity
- Several approaches proposed
 - From “based on expert opinions” to application of some algorithms or a number of factors that influence the development cost (e.g., size)
- Context of small/medium software companies that do not have data to build their own models
 - Use of publicly available dataset (e.g., PROMISE)
 - Cross-company models

Issues

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- Accuracy of Cross-company models can be significantly worse than those of estimation models built employing datasets collected by the company from past projects (within-company models)
 - Different characteristics, different contexts
- A solution: selecting features and observations
- Proposed selection strategies require:
 - datasets with many observations
 - availability of some data of the single company
 - sophisticated learning strategies

Focus of the paper

Motivation

- *availability of some data of the single company*
- *sophisticated learning strategies*

- How small/medium company can deal with these issues?
- Our goal was to identify an easy strategy based on opinions of the company expert managers when data are not available

Study Design

- Experts select data taking into account characteristics of public datasets

Study Results

- Proposal: use of Delphi approach to reach consensus among experts involved in the selection

Final remarks

- Linear regression to build the prediction models

Research question

Motivation

- RQ: Are the effort predictions obtained with CCexpert better than those achieved with CCoriginal?

Study design

- CCoriginal: exploits all the observations of the publicly available dataset
 - $obs_1, obs_2, \dots, obs_n$ in the cross-company dataset

Study Results

- CCexpert: exploits only data selected by the selection strategy based on expert opinion
 - $obs_i, obs_z, \dots, obs_t$ (where $t \leq n$)

Final remarks

Training set

Motivation

Study Design

Study Results

Final remarks

- The dataset employed to build the cross-company models was selected from the PROMISE repository (info extracted from ISBSG)
 - COSMIC
 - information on the development process

Test set

Motivation

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- Data provided by an Italian software company
 - Specialized in the design, development, and management of solutions for Web portals, enterprise intranet/extranet applications and GIS
 - certified ISO 9001:2000, and also a certified partner of Microsoft, Oracle, and ESRI
- The company provided us a set of 25 applications
 - e-government, e-banking, Web portals, and Intranet applications developed by exploiting a wide range of Web-oriented technologies, such as J2EE, ASP.NET, etc.

Data Selection Strategy

Motivation

- *ALM: selection of observations, i.e., projects, from the cross-company dataset that have characteristics more close to the ones developed by the single company*

Study Design

- Delphi approach: questions in two rounds

Study Results

- an anonymous summary of all judgments is provided by a facilitator after each round
- experts are encouraged to revise their previous answers taking into account the opinions of the other participants
- process can be stopped according to a predefined stop criterion (e.g., achievement of consensus, stability of results)

Final remarks

Estimation technique, validation method, and evaluation criteria

Motivation

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- Simple Linear Regression (SLR)
 - $EFF = \alpha * CFP + b$
- Hold-out cross validation
- Mean of the Absolute Residuals (MAR)
- T-/Wilcoxon Test to verify ARs obtained with CCexpert are significantly less (i.e., better) than ARs achieved with CCoriginal (a probability of 5% of committing a Type-I-Error)
- Vargha and Delaney's A12 statistics as non-parametric effect size measure

Threats to validity (1)

Motivation

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Construct validity

- The collection of the information to determine size measures and actual effort represents a crucial aspect

Conclusion validity

- Chosen evaluation criteria allowed assessing in an objective way the effectiveness of the predictions
- We carefully applied the estimation methods and the statistical tests, verifying all the required assumptions

Threats to validity (2)

Motivation

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Reliability validity

- all the information exploited in our study to select the observations in the training set (used to build the effort estimation model) are available on the web, i.e., PROMISE repository.

External validity

- we are aware that we considered one dataset to test the approach and more tests should be exploited to generalize the results.

Cross-Company Dataset Selection

Motivation

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- three managers of the selected company were involved in the application of the Delphi, two times:
 - to decide what variables of the cross-company dataset should be considered
 - to select the observations according to the chosen features
- Variables selected: Primary Programming Language (PPL); Language1; DBS1; Architecture; OS1
- Observations selected: 8 out of 42

Cross-Company Dataset Selection

Motivation Study Design Study Results Final remarks	Variable	Description
	PPL	The primary language used for the development: JAVA, C++, PL/1, Cobol etc.
	Language1	The primary technology programming language used to build or enhance the software, i.e., that used for most of the build effort
	OS1	The primary technology operating system used to build or enhance the software, i.e., that used for most of the build effort
	Architecture	A derived attribute for the project to indicate if the application is: Stand alone, Multi-tier, Client server, Multi-tier with web public interface
Study Results	DBS1	The primary technology database used to build or enhance the software (i.e. that used for most of the build effort), otherwise (if known) it is whether the project used a DBMS
Final remarks		

Cross-Company Dataset Selection

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obs	CFP	EFF	PPL	Language1	OS1	Architecture	DBS1
29331	210	204	Java	C++ Java	UNIX Windows NT	Multi-tier	Oracle
25081	237	600	C++	Java	Windows	Multi-tier with web public interface	MySQL
29471	61	669	Java	Java	Linux	Multi-tier with web public interface	MySQL
30658	5	784	ABAP	Java	Windows	Multi-tier with web public interface	MySQL
31895	82	903	C#	Java	Windows XP	Client server	SQL Server
27505	118	1488	Script Language	Java	Linux	Multi-tier with web public interface	MySQL
29311	643	4224	Java	Java	Windows 2000	Client server	Oracle
29310	441	47493	ASP.Net	Java 2EE	Windows XP	Client server	Oracle

Cross-Company Dataset Selection

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obs	CFP	EFF	PPL	Language1	OS1	Architecture	DBS1
29331	210	204	Java	C++ Java	UNIX Windows NT	Multi-tier	Oracle
25081	2						QL
29471							QL
30658							QL
31895							Server
27505	1						QL
29311	6						le
29310	4						le

- the values of the selected variables are close to the ones that characterize the projects that usually the company develop
- the company employs SUN J2EE or Microsoft .NET technologies for the software development.
- Oracle has been the most commonly adopted DBMS. However, also SQL Server and MySQL are employed.

Cross-Company Dataset Selection

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- the facilitator observed that many observations were not considered since they
 - do not include information about Architecture and DBS1
 - or do not specify the value for Language1
 - specify values of PPL and Language1 not close
 - COBOL

Model Construction and Validation

Motivation

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Study Results

Final remarks

- Verification of linear regression assumptions
 - log transformation of the variables
- Models transformed back to the original raw data scale:
 - $EFF = 738 * CFP^{0.228}$ (CCoriginal)
 - $EFF = 161 * CFP^{0.456}$ (CCexpert)

Model Construction and Validation

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Study Design

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Model	MAR
CCoriginal	701
CCexpert	467

Statistically significant difference
(p-value <0.001)

Model Construction and Validation

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Model	MAR
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Final remarks

The effort predictions obtained with CCexpert are better than those achieved with CCoriginal

Main findings

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Study Design

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Final remarks

- Proposed strategy has provided interesting results for the software company involved in our study
 - can be improved by automating the selection of variables
- (Point of view of practitioners) *a widely recognized approach (i.e., Delphi) can allow to improve significantly the effort predictions, with a minimal effort, in a context of small/medium company which does not have data available from past developed projects*

Future work

Motivation

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- factors such as the application area of the projects, the size of the available development group, or performance and security requirements were not considered
 - verify these decisions in other (similar) contexts
- we employed only a functional size measure, i.e., COSMIC, since the company measured the projects made available for the study with this measure
 - consider other and more measures
- verify the managers' opinions of the cost-benefit of the approach, by performing a qualitative research study